

The non-unity of differential object marking: Flagging versus indexing

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Abstract

This paper shows that differential object marking (DOM) is not a unitary cross-linguistic phenomenon, and observes that its two subtypes, differential object flagging (DOF) and differential object indexing (DOI), probably have rather different explanations. The cross-linguistic patterns found with both subtypes of DOM can be described by the scales of referential prominence (animacy, specificity, and so on), but DOI is like differential subject indexing (DSI) in that it conforms to the prominent indexing universal, while differential subject flagging (DSF) shows the mirror-image pattern of DOF: Flagging is preferred when the subject is not prominent on one of the prominence scales. The paper provides examples from a wide variety of languages showing that indexing behaves rather differently from flagging. Thus, DOM is not a unified phenomenon, as was already noted by Siewierska (2004) and some earlier authors.

Keywords: differential object marking, indexing, flagging, referential prominence, coding efficiency.

1. Introduction

This paper compares differential object flagging (DOF) with differential object indexing (DOI) in the world's languages, as well as with two kinds of differential

subject marking (DSF and DSI).¹ It notes that the two types of differential marking are more different than has sometimes been suggested, and it argues (following Just 2024) that different explanations should be given for the generalizations obeyed by differential flagging and differential indexing. The two kinds of differential marking are thus not a unified phenomenon from a general-comparative perspective.

Differential object marking (DOM) has long been prominent in Romance linguistics and beyond (e.g. Bossong 1991; Aissen 2003; Kabatek et al. 2021). It describes a situation in which two classes of patient objects are marked in different ways depending on their intrinsic properties (such as animacy) or on the discourse context (e.g. definiteness or topicality). For example, Spanish uses its accusative marker *a* primarily with animate patient objects, as illustrated in (1a-b).

(1) Spanish (Fábregas 2013: 1): differential object flagging (DOF)

a. *Encontré un problema.*

I.found a problem

‘I found a problem.’

b. *Encontré a un superviviente.*

I.found ACC a survivor

‘I found a survivor.’

Similarly, Swahili uses its object person index (*ki-* in example (2b)) only when the nominal object is topical, as illustrated in (2a-b).

(2) Swahili (Mursell 2018: 427; 433): differential object indexing (DOI)

a. *Mwanamke a-li-vunja kikombe.*

woman(G1) SBJ.G1-PST-break cup(G7)

‘The woman broke a cup.’

b. *Mwanamke a-li-ki-vunja kikombe.*

woman(G1) SBJ.G1-PST-OBJ.G7-break cup(G7)

‘The woman broke the (aforementioned) cup.’

The Spanish type of object marking was generally known as “prepositional accusative” in Romance languages before the 1990s, but it is actually found widely in the world’s languages, and it may also be manifested by a special case affix. Following Moravcsik (1978), Bossong (1982; 1985) called the phenomenon differential object marking, and he did not discuss differential marking on the verb (as in Swahili) in his early work. But in subsequent research on Romance languages, both differential object marking on the nominal and differential object marking on the verb have been discussed widely, for example in Romanian:

¹ The terms DSF (differential subject flagging) and DSI (differential subject indexing) are introduced here, as special instances of differential subject marking (DSM) (see below in §3). The terminology is thus completely parallel: DOM (differential object marking) has the two subtypes DOF and DOI, and DSM has the two subtypes DSF and DSI (see Figure 1 in §7 for a summary of the terminology).

(3) Romanian (Irimia 2020: 429)

- a. *Maria a văzut o stea.*
 Maria has seen a star
 ‘Maria saw a star.’
- b. *Maria l-a văzut pe Ion.*
 Maria 3SG.OBJ-has seen ACC Ion
 ‘Maria saw Ion.’

A similar pattern is found in Syriac (an ancient Semitic language), e.g.

(4) Syriac (Coghill 2014: 358)

- a. *Bnā baytā.*
 built.3SG.SBJ house
 ‘He built a/the house.’
- b. *Bnā-y(h)i (l-)baytā.*
 built.3SG.SBJ-3SG.OBJ (ACC-)house
 ‘He built the house.’

In order to be clear about the two situations, it is useful to have different terms for the two subtypes of differential object marking in a cross-linguistic perspective: For the first type, we can say *differential object flagging* (DOF), using *flag* as a cover term for case and adpositional marking (Haspelmath 2019b), and *differential object indexing* (DOI), using *index* as a cover term for agreement marker and doubling clitic (Haspelmath 2013).

In the Romanian example in (3), we see both differential object flagging (with the preposition *pe*) and differential object indexing (prefix *l-*) occurring under similar conditions especially when the object is definite. It is therefore not far-fetched to suspect that the two phenomena should be identical in some abstract sense, and that they should be explained in the same (or an analogous) way. This idea is formulated in (5).

(5) **Presumption of DOM unity**

Differential object flagging and differential object indexing are manifestations of the same underlying or motivating factor.

This presumption can be found prominently in the earlier literature (where object indexing was typically called “agreement”). For example, Morimoto (2002: 294) says:

“In this paper, I present data from Bantu languages, which are primarily head marking, and argue that DOM in case marking languages and previously observed variation in the use of object agreement in some Bantu languages are one and the same phenomenon conditioned by the single generalization in (1).

- (1) The higher in prominence a direct object the more likely it is to be overtly case marked – where the dimensions along which prominence is assessed include animacy ... and definiteness ...”

Similarly, Dalrymple & Nikolaeva (2011: 1-2) say that they “understand DOM as covering both agreement and case-marking (case or adpositional marking on the object)”, and they provide analogous explanations for them in their book. Their view is echoed by Kalin (2018: 113):²

“On an inclusive conception of DOM, which I adopt here, marking may take the form of case (e.g., Persian, Turkish), an adposition (e.g., Spanish, Hindi), agreement (e.g., Swahili, Ostyak), or clitic doubling (e.g., Macedonian, Catalan).”

However, we will see in this paper that there are good reasons to reject the presumption of unity of differential object marking, and to explain its two subtypes DOF and DOI in different ways. In the next two sections, we will see (following Just 2024) that not only object indexing (§2), but also subject indexing (§3) is driven by referential prominence when it is variable or differential. (I call this the “prominent indexing universal” in §2.1 below.)

By contrast, the variable or differential occurrence of flagging with objects or subjects (§4) obeys a different cross-linguistic generalization: Deviations from usual role-reference associations tend to get special marking (as explained in Haspelmath (2021b) in terms of efficient coding; see §5). We will see that these generalizations also extend to flagging and indexing elsewhere: Role-reference associations are also responsible for differential place flagging, and the prominent indexing universal also extends to indexing of recipients and adpossessors (§6).³ Finally, I note that the crucial contrast between differential flagging and differential indexing that is highlighted here is not a new insight but was noted by several researchers in earlier works (§7).

I should not here that in this paper, I am concerned exclusively with argument marking by flags and indexes, and I say nothing about word order or binding relations, which have played an important role in the literature. Differential object marking, in particular, is often accompanied by object fronting of some kind (e.g. in Sakha, as highlighted by Baker & Vinokurova 2010). However, there are many languages where differential object marking is quite independent of word order (and many languages where objects occur in different positions independently of object flagging), so I am assuming that it makes sense to adopt a perspective in which the generalizations are treated independently.

² See also Bárány & Kalin (2020) for a perspective similar to Kalin’s, and Aissen (2003: 474) for an endorsement of Morimoto’s approach. One reviewer suggests that the “presumption of DOM unity” is not the majority view in the field, but it is surely not merely a strawman. The term “DOM” used for both flagging and indexing could certainly invite the conclusion that the phenomenon has a uniform explanation, so it is worth examining the presumption of unity.

³ The generalizations also extend to scenario-based differential coding, as it is attested in Spanish (see García García (2014) and Kabatek (2016), who emphasize that differential object marking of an inanimate nominal is possible in Spanish when the subject is inanimate; see Haspelmath 2021b: §6.4).

2. Differential object indexing (DOI) and the prominent indexing universal

As noted in the introductory section, there are two types of DOM: differential object flagging (DOF) and differential object indexing (DOI).⁴ In Romance linguistics, these two subtypes have also been called “flagging DOM” and “indexing DOM” (e.g. Melis 2018; Gerards 2023). DOF is much better known than DOI, and we will get back to it in §4 below. In the next two sections, the focus will be on differential indexing, both object indexing (DOI, §2) and subject indexing (DSI, §3).

2.1. Referential prominence

In many or most cases, differential argument indexing is conditioned by *referential prominence*. We saw above that Romanian and Syriac show indexing of definite objects but not necessarily of indefinite inanimate objects, and Swahili indexes topical objects but not nontopical objects. This is a very common pattern and can be generalized as in (5), following much earlier work on the topic (e.g. Siewierska 2004: 149; Iemmolo 2011: 268; Haig 2018; Just 2022; 2023; 2024).

(6) The prominent indexing universal (preliminary version)

When argument indexing is differential and is conditioned by referential prominence, all languages prefer indexing of referentially prominent arguments.

In other words, what we (almost) never find is differential indexing where the indexing occurs with referentially non-prominent argument and fails to occur with prominent arguments.

By *argument indexing*, I primarily mean person-number indexing (following the terminology in Haspelmath 2013), i.e. argument marking by bound person-number forms on the verb, in addition to an argument nominal.⁵ I will mostly consider verbal person indexes, but §6 below briefly considers adpossessionive person indexing on possessed nouns, and I also include number marking on verbs in this paper. The notion of indexing does not distinguish between “agreement” and “pronouns”, or between marking by “affixes” or “clitics”. In Romance languages, object indexes have often been called “clitics”, and when they cooccur with an object nominal, the construction has been called “clitic doubling” (e.g. Anagnostopoulou 2006; Poletto & Pinzin 2024). I will not use this terminology here, but will use the term *indexing* as it is now established in much cross-linguistic work (e.g. Croft 2022; Creissels 2024).⁶

⁴ The term DOI (differential object indexing) was first used by Iemmolo (2011) and Iemmolo & Klumpp (2014). A synonymous term is “differential P indexing” (Just 2024), and a somewhat broader concept is “conditional indexing” (Walker 2024). An earlier term was “differential object agreement” (Lazard 2001: 880; “accord objectal différentiel”), which was adopted by Coghill (2014) (who contrasted DOF (differential object flagging) with DOA (differential object agreement)).

⁵ Person-number indexing may of course also occur in the absence of a conominal (e.g. Romanian *Maria l-a văzut* ‘Maria saw him’, cf. (3b)). This usage is not considered here, even though it may of course also show variability.

⁶ Linguists have often suggested or assumed that affixes and clitics behave differently with respect to the “agreement” vs. “pronoun” distinction (see, e.g., Anagnostopoulou 2016 for discussion). But there does not seem to be any generally established view on this, and for

Now what exactly is a referentially prominent argument? This notion is defined in terms of one (or more) of the six prominence scales in (7)-(8) (e.g. Aissen 1999; Siewierska 2004: 149; Lockwood & Macaulay 2012; Haspelmath 2021b: 127-128).

(7) scales of inherent prominence

person scale:	locuphoric (1st/2nd) > aliophoric (3rd person)
full nominality scale:	person form (independent or index) > full nominal
animacy scale:	human (> animal) > inanimate

(8) scales of discourse prominence

specificity scale:	definite (> specif. indefinite) > indefinite nonspecific
givenness scale:	discourse-given > discourse-new
focus scale:	background > focus ⁷

Thus, the prominent indexing universal in (6) means that if an argument is higher on one of these scales, it is more likely to be indexed than if it lower. For example, human arguments are more likely to be indexed than inanimate arguments, and discourse-given arguments are more likely to be indexed than discourse-new arguments.

The universal in (6) is phrased in terms of “languages preferring” A over B, which is meant as an abbreviation of the claim that A is more likely to occur than B. Alternatively and more precisely, one can formulate the claim in terms of an implicational universal:

(9) **The prominent indexing universal** (definitive version)

If a referentially less prominent argument is indexed in a language, the argument types higher on one of the prominence scales are also indexed.

This implicational universal allows three language types: Those with no indexing, those with uniform (non-differential) indexing, and those with differential indexing where only the more prominent arguments are indexed.

In the following subsections, we will see examples of differential object indexing conditioned by various prominence factors. We start with inherent

the comparative purposes of this paper, I treat all person indexes (whether affixes or clitics) alike, and I do not distinguish between cross-indexing and gramm-indexing here (see Haspelmath 2013: §4.2-3). A reviewer objects that this is an “unacceptable simplification”, but I am actually agnostic as to how language-particular phenomena should be analysed, and this paper is not incompatible with generative works like Yuan (2021), which make a clear distinction between agreement and clitics. All I am saying is that if one adopts the perspective of Haspelmath (2013), using “indexing” as a comparative concept, certain testable generalizations about differential object indexing can be stated.

⁷ For “discourse prominence”, the terminology is not intuitive in all respects. In the present context, focus has lower “referential prominence” than background, but of course it has higher “prominence” in other senses of the word. The term “referential prominence” is strictly defined in terms of the scales in (7)-(8), and I leave it open what they might have in common. Their main purpose here (and in Haspelmath 2021b) is to serve as comparative concepts for formulating universal generalizations.

prominence (pronominality, animacy) and then consider discourse prominence (definiteness, topicality).⁸

2.2. Pronominality-conditioned DOI

In some languages such as Spanish, object indexing occurs with personal pronoun objects but not with full nominals. (See also Kayne 2000: 164-165 for French.)

(10) Spanish

- a. *Veó a Pedro.*
 I.see ACC Pedro
 ‘I see Pedro.’
- b. *Lo veó a él.*
 3SG.OBJ I.see ACC him
 ‘I see him.’

2.3. Animacy-conditioned DOI

In some languages such as Oneida, Bunaq and Pangkhua, object indexing occurs with animate objects but not with inanimate objects.

(11) Oneida (Iroquoian; Koenig & Michelson 2015: 6)

- a. *k-nú'weh-se?*
 1SG.A-like-HAB
 ‘I like it.’
- b. *khe-nú'weh-se?*
 1SG>3-like-HAB
 ‘I like her (or him, or them).’

(12) Bunaq (Timor-Alor-Pantar; Schapper 2022: 125)

- a. *Markus zo poi.*
 Markus mango choose
 ‘Markus chose a mango.’
- b. *Markus zap go-poi.*
 Markus dog 3.OBJ-choose
 ‘Markus chose a dog.’

⁸ In the literature on Romance languages, the two types of prominence factors are sometimes called “inherence factors” and “reference factors” (e.g. Bossong 1991: 158-160), but here I use the labels *inherent prominence* and *discourse prominence* (as subtypes of referential prominence). Instead of “referential prominence”, some authors have said “topicality/topicworthiness” (e.g. Siewierska 2004: 149), or “salience” (e.g. Croft 2022), and instead of “scale”, many authors say “hierarchy”. All of these differences are purely terminological.

(13) Pangkhua (Kuki-Chin, Trans-Himalayan; Akter 2024: 272)

- a. *Ramŋai par t̪amkan ə=mu?*
 Ramngai flower many 3SG.SBJ=see
 ‘Ramngai saw many flowers.’
- b. *Ramŋai vɒk t̪amkan ə=mu=ei.*
 Ramngai pig many 3SG.SBJ=see=OBJ.PL
 ‘Ramngai saw many pigs.’

2.4. Definiteness-conditioned DOI

In some languages such as Bulgarian, Mauwake (Trans-New Guinea) and Ho-Chunk (Siouan), object indexing occurs with definite objects but not with indefinite objects.

(14) Bulgarian (Compensis 2022: 30)

- a. *Kuče-to goni kotka.*
 dog-DEF.N chases cat-DEF.F
 ‘The dog chases a cat.’
- b. *Kuče-to ja goni kotka-ta.*
 dog-DEF.N 3SG.F.OBJ chases cat-DEF.F
 ‘The dog chases the cat.’

(15) Mauwake (Trans-New Guinea; Berghäll 2015: 97; 216)

- a. *Bom=iya kateres=iya fuurk-a-mik.*
 bomb=COM cartridge=COM drop-PST-3PL
 ‘They dropped bombs and cartridges.’
- b. *Emeria wia=amukar-e-k.*
 woman 3PL.ACC=scold-PST-3SG
 ‘She scolded the women.’

(16) Ho-Chunk (Helmbrecht 2021: §6.2.3)

- Wagujé(=ra) (wa-)išé-re.*
 moccasin(=DEF) (3PL.OBJ-)take.away-3PL.SBJ
 ‘They took (the) moccasins away.’

2.5. Topicality-conditioned DOI

In some languages, object indexing occurs with topical objects but not with nontopical objects. Example (17) shows this for gender-number agreement expressed in the verb stem in the Italian variety of Sanvalentine.

(17) Sanvalentine (Abruzzo) (D’Alessandro 2017: 7)

- a. *Ajə cciosə li pellistrə.*
 I.have killed.M.SG the chickens.M.PL
 ‘I have been killing CHICKENS.’

- b. *Ajə ccisə li pellistrə.*
 I.have **killed.M.PL** the chickens.M.PL
 ‘I have KILLED the chickens.’

In the following examples from Tundra Nenets (Uralic), the preceding context shows that the object is non-topical in (18a), but topical in (18b). The agreement suffix *-da* occurs only in the latter case.

(18) Tundra Nenets (Dalrymple & Nikolaeva 2011: 128)

- a. (‘Whom did Vanya hit?’)
Wanya Pyetya-m ladə°. (**ladə°da*)
 Vanya Petya-ACC hit.3SG.SBJ
 ‘Vanya hit PETYA.’
- b. (‘What did a/the man do to a/the reindeer?’)
Xasawa ti-m xada°-da. (**xada°*)
 man reindeer-ACC killed.3SG.SBJ-3SG.OBJ
 ‘A/the man KILLED a/the reindeer.’

Similarly, in Swahili, the object index cannot be used when the nominal object is focused, regardless of its definiteness (Mursell 2018: 433).

(19) Swahili

- a. *Mwanamke a-li-(*ki-)vunja nini?*
 woman(G1) SBJ.G1-PST-break what
 ‘What did the woman break?’
- b. *Mwanamke a-li-(*ki-)vunja kikombe.*
 woman(G1) SBJ.G1-PST-OBJ.G7-break cup(G7)
 ‘The woman broke a/the cup.’

3. Differential subject indexing (DSI)

In addition to differential object marking (DOM), languages may also exhibit differential subject marking (DSM; see de Hoop & de Swart 2009).⁹ Here, too, there are two subtypes: differential subject flagging (DSF) and differential subject indexing (DSI). We first look at DSI, in the current section, before moving on to differential flagging (DOF and DSF) in §4 below. Similarly to DOI, DSI is typically conditioned by referential prominence, as initially illustrated by (20) from Awing: When the subject is topical, as in (20b), it may be indexed by the bound form *a*, but when it is focused, as in (20a), it may not be indexed.

⁹ A subject is defined as an A-argument of a transitive clause or an S-argument of an intransitive clause (following Dixon 1994: 113). Some works focus specifically on “differential marking of the A-argument” (e.g. Just 2024), and it is certainly interesting to ask how S-arguments and A-arguments differ with respect to differential marking. But here I lump them together for simplicity.

(20) Awing (Bantoid; Fominyam & Georgi 2021: 84)

- a. *Wə (*a) nə n-náŋnə məʒíə əzoonə.*
 who(3SG) PST N-cook food yesterday
 ‘Who cooked the food yesterday?’
- b. *Alombah (a) nə n-náŋnə məʒíə əzoonə.*
 Alombah 3SG PST N-cook food yesterday
 ‘Alombah cooked the food yesterday.’

Next we will see a number of additional examples of DSI conditioned by referential prominence. All of them obey the same universal generalization as DOI, the prominent indexing universal of §2.1.

3.1. DSI conditioned by inherent prominence

Differential subject indexing may be conditioned by animacy, as in Nez Perce (a Sahaptian language of North America), where only animate nominals trigger suffixal number agreement on the verb (Deal 2016: 318). This applies both to subject indexing and object indexing.

(21) Nez Perce (Deal 2016: 318)

- a. *Lep-ít cepéepy’uǰtin’ hii-we-s ʔiniit-pe.*
 two-SUFF pie 3.SBJ-be-PRS house-LOC
 ‘Two pies are in the house.’
- b. *Lep-ít pícpic hi-w-s-íix ʔiniit-pe.*
 two-SUFF cat 3.SBJ-be-PRS-S.PL house-LOC
 ‘Two cats are in the house.’

The situation is very similar in Garifuna (an Arawakan language of Central America). According to Barchas-Lichtenstein (2012: §4.2.1), only animate nominals may trigger verbal plural marking: The plural marker *-tiyan* occurs with animates such as ‘dogs’, as in (22b), but not with inanimates such as ‘flowers’ or ‘balls’. Again, this applies to objects (as in 22) and subjects (as in 23) in the same way.

(22) Garifuna (Barchas-Lichtenstein 2012: 167-168)

- a. *Éiha n-umu-ti óunli.*
 see 1SG-AUX-3M dog
 ‘I saw the dog.’ (*‘I saw the dogs.’)
- b. *Éiha n-umu-tiyan óunli.*
 see 1SG-AUX-3PL dog
 ‘I saw the dogs.’
- c. **Éiha n-umu-tiyan fulúri.*
 see 1SG-AUX-3PL flower
 (‘I saw the flowers.’)

(23) Garifuna (Barchas-Lichtenstein 2012: 167-168)

- a. *Brí-ti báalu.*
be.good-3M ball
'The ball is good' or 'The balls are good.'
- b. **Brí-tiyan báalu.*
be.good-3PL ball
('The balls are good.')

A better-known language in which animacy plays a role is Turkish (e.g. Alexiadou 2019: 141), where inanimate subject nominals do not trigger the plural agreement suffix *-lar* on the verb, while animate subjects do (even if they are indefinite, though this is obligatory only when they are definite).

(24) Turkish

- a. *Anahtar-lar kapı-lar-ı aç-tı(-*lar).*
key-PL door-PL-ACC open-PST(-3PL)
'Keys opened the doors.'
- b. *Çilingir-ler kapı-lar-ı aç-tı(-lar).*
locksmith-PL door-PL-ACC open-PST(-3PL)
'Locksmiths opened the doors.'

DSI may also be conditioned by pronominality, as illustrated by Welsh. As is well-known, only personal pronouns (but not full nominals) trigger person marking on the verb in this language:

(25) Welsh (Borsley et al. 2007: 199)

- a. *Gwel-odd y bechgyn ddraig.*
saw-3SG the boys dragon
'The boys saw a dragon.'
- b. *Gwelo-n nhw ddraig.*
saw-3PL they dragon.
'They saw a dragon.'

3.3. DSI conditioned by discourse prominence

A better-known phenomenon is the role of topicality for triggering indexing. Thus, in English and French, postverbal (i.e. nontopical) plural subjects do not (need to) occur with plural verb forms, as illustrated by (26)-(27) (Lambrecht 2000: 642-643).

(26) English

- a. *The three women **are** IN THE ROOM.*
- b. *There's three WOMEN in the room.*

(27) French

- a. *Les trois femmes sont venues.* (*est venu)
 the three women are come.FEM.PL (is come)
 ‘The three women CAME.’
- b. *Il est venu trois femmes.* (*sont venues)
 it is come.MASC.SG three women (are come)
 ‘There came three WOMEN.’

While DOI is often seen as “additional indexing” (or “doubling”), differential subject indexing (DSI) tends to be taken as “absent indexing”, and the presence of indexing is seen as the default. For example, Lambrecht’s (2000: §3.2) refers to the phenomenon in (26b) and (27b) as “suspended subject-verb agreement”. In the present context, this difference is irrelevant, as there is no directionality in generalizations such as (9).

A language with reduced indexing for discourse-new arguments is Yimas (of the Sepik region in Papua New Guinea). According to Foley (1991: 232), arguments that “convey new information” or refer to characters “just being introduced in the discourse” need not be indexed on the verb. In the two contrastive examples in (28), (a) shows full indexing on the verb (for both A and P arguments), while (b) shows no indexing because the S argument is new information.

(28) Yimas (Foley 1991: 228; 233)

- a. *Kacmpt payum ya-mpu-yamal-wat.*
 canoe.PL(G8) man.PL(G1) PL.G8.P-3PL.A-carve-HAB
 ‘The men usually carve the canoes.’
- b. *Numn-mat Kampramanan wapal-k.*
 villager-PL Kampramanan climb-IRR
 ‘The villagers climbed Kampramanan.’

Finally, let us briefly consider long-distance agreement in Tsez, a Dagestanian language whose verbs show gender-number agreement with the absolutive argument. In some kinds of subordinate clause, the matrix verb can agree with the absolutive argument of the subordinate clause, as in (29a), where *b-iy-xo* ‘knows’ shows the class 3 prefix *b-*, agreeing with *magalu* ‘bread’. This agreement is optional, so that the verb may also show default agreement (*r-*), as in (29b). However, long-distance agreement is not possible when the absolutive argument is focal, as in (29c).

(29) Tsez (Polinsky & Comrie 1999: 120, 124)

- a. *Eni-r [už-ā magalu b-āc'-ru-λin] b-iy-xo.*
 mother-DAT boy-ERG bread.ABS(CL3) CL3-eat-PST-COMP CL3-know-PRS
 ‘The mother knows that the boy ate bread.’
- b. *Eni-r [už-ā magalu b-āc'-ru-λin] r-iy-xo.*
 mother-DAT boy-ERG bread.ABS(CL3) CL3-eat-PST-COMP CL4-know-PRS
 ‘The mother knows that the boy ate bread.’

- c. *Eni-r* [už-ā *magalu-kin* *b-āc'-ru-λin*] **b/r-iy-xo*.
 mother-DAT boy-ERG bread.ABS(CL3)-FOC CL3-eat-PST-COMP CL3/CL4-know-PRS
 ‘The mother knows that the boy ate BREAD.’

So again, discourse status plays a role in allowing indexing (or agreement), and again, focal status disfavors agreement. The Tsez type of agreement (long-distance agreement) is quite different from the cases seen earlier, but the conditioning is quite parallel.¹⁰

3.4. Summary on differential indexing

In §2 and §3 of this paper, we saw that both differential object indexing (DOI) and differential subject indexing (DSI) obey the prominent indexing universal of (9) above, repeated here:

(9) The prominent indexing universal

If a referentially less prominent argument is indexed in a language, the argument types higher on one of the prominence scales are also indexed.

So far, the evidence for this universal is only impressionistic, and cannot of course be established by citing a few examples from a few languages, as I did in these two sections. It may be worth doing a systematic comparative study, though this is not easy as restrictions on indexing (or agreement) constructions are not always fully described in reference grammars. However, the proposed universal has been known for some time,¹¹ and if clear counterevidence had come to light, it would probably have become known. In the next section, we will see that the generalization is different for differential flagging.

4. Differential object flagging (DOF) and differential subject flagging (DSF)

In the past literature, most work on differential argument marking, and more on specifically differential object marking, has been devoted to DOF (differential object flagging via adpositions or case). This phenomenon is very well-known, and it was illustrated for Spanish in (1) above. Two more examples are given in (30)–(31).

¹⁰ For further discussion of “loss of agreement” with focal subject arguments, see also Mereu (1999), Siewierska (2004: 159–161), Malchukov & Ogawa (2011: §3.3), and Just (2024: §3).

¹¹ For example, Woolford (1999: 203) observed that “it is well-known that objects whose features are high on one of the animacy/topicality hierarchies (e.g. human, specific, first person, etc.) are more likely to trigger agreement”, and Barlow (1999: 192) noted that “in general, high definiteness induces agreement, whereas low definiteness is associated with the absence of agreement” (and similarly for animacy). The work by these two authors from different researcher communities was widely read.

(30) Turkish (von Heusinger et al. 2019: 146)

a. *Ben elma ye-di-m.*

I apple eat-PST-1SG

‘I have eaten an apple.’ (non-specific object nominal)

b. *Ben elma-yı ye-di-m.*

I apple-ACC eat-PST-1SG

‘I ate the apple.’ (specific object nominal)

(31) Zhoutun Mandarin (Zhou 2022: 117, 24)

a. 你苹果吃了么?

Ni phĩkur tʃhi=lɔ mɿ?

you apple eat=PFV Q

‘Did you eat apples?’

b. 苹果哈我一个吃了。

Phĩkur=xa ŋɿ i=kɿ tʃhi=lɔ.

apple=ACC I one=CLF eat=PFV

‘An apple I ate.’

That differential object marking is typically conditioned by high referential prominence has long been known, and was first stated clearly by Moravcsik (1978), in the context of Joseph Greenberg’s universals project at Stanford University. A quotation (with some irrelevant detail omitted) is given in (32).¹²

(32) **The differential object flagging universal** (Moravcsik 1978: 283)

“If there is any semantic difference between an accusative marking and ... a nominative marking ..., this semantic difference will be related either to definiteness, or to animacy, or humanness ..., with the accusative ... marking the more definite (rather than the less definite), the animate or human (rather than the inanimate or non-human) ... noun phrase.”

But in addition, languages may also exhibit differential subject flagging, exemplified here by Central Tibetan, where only focused A-arguments get an ergative case suffix.

(33) Central Tibetan (Tournadre 1995: 264)

a. *khōng khāla’ so-kiyo:re’*

he food make-IPFV.GNOM

‘He prepares the meals.’ (no flag on topical A-argument)

b. *khōng-ki’ khāla’ so-kiyo:re’*

he-ERG food make-IPFV.GNOM

‘HE prepares the meals.’ (ergative flag on focused A-argument)

¹² See also my 2022 blogpost about the early history of this generalization: <https://dlc.hypotheses.org/2764>

In general, differential subject marking tends to be conditioned by person, animacy or focus status, as also illustrated by Punjabi (with ergative marking only on 3rd person subjects), Mangarrayi (with ergative marking only on inanimate subjects), and Yali (New Guinea, with ergative marking only on subjects with low discourse prominence such as focus) (see Haspelmath 2021b: §4.2 for more on this).

(34) Punjabi (Chandra & Kaur 2017: 26)

a. *Maĩ(*-ne) rotti khaadd-ii.*
I(-ERG) bread(F) eat-F.SG
'I ate bread.'

b. *Munde-ne rotti khaadd-ii.*
boy-ERG bread(F) eat-F.SG
'The boy ate bread.'

(35) Mangarrayi (northern Australia; Merlan 1982: 61)

a. *Buy? ñan-wu-na ŋala-bugbug?*
show 3SG>2SG-AUX-PST.PUNCT F.NOM-old.woman
'Did the old woman show you (to him)?'

b. *Na-ŋugu ñim ŋan-ga-ŋiñ.*
ERG-water submerge 3SG>1SG-AUX-PST.PUNCT
'Water covered/submerged me.'

(36) Yali (Trans-New Guinea; Riesberg 2018: 19)

a. *Hiyap tu(=en) wanggun fam wam wat-tuk.*
woman DEM(=ERG) stick with pig hit-PROG
'The woman is hitting a pig with the stick.'

b. *Wam itno hiyap tu=en ambol=mu wat-tuk.*
pig DET woman DEM=ERG back=LOC hit-PROG
'The WOMAN is hitting the pig on the back.'

Now the key observation is that the overall generalization for differential flagging of core arguments is not uniform: While both DOF and DSF are often conditioned by referential factors of the kind seen in the prominence scales in (7)-(8), the directionality of the scalar generalizations is the opposite. We can formulate the overall generalization as in (37), restricting attention to accusative and ergative flagging.¹³

(37) **The core argument flagging universal**

(a) If in a language, a referentially less prominent P-argument is accusative-flagged, the argument types higher on the scale are also flagged. (b) If a

¹³ Differential subject flagging could in principle also be differential nominative flagging, but since nominative flags are so rare, this does not play a significant role and is left out here for simplicity.

referentially more prominent A-argument is ergative-flagged, the argument types lower on the scale are also flagged.

This means that there are languages with accusative marking everywhere (e.g. Quechua), and languages with accusative marking only on prominent argument nominals, but not the opposite. Conversely, there are languages with ergative marking everywhere (e.g. Lezgian), and languages with ergative marking only on non-prominent argument nominals, but not the opposite

The pattern for P-arguments in (a) is thus the mirror image of the pattern A-arguments in (b), as noted by Dixon (1994: 85) and other scholars.¹⁴ Moreover, as noted in Haspelmath (2021b), the generalization in (37) can be extended to the T-argument and the R-argument of ditransitive clauses: The T-argument behaves like the P-argument in its flagging preferences, while the R-argument behaves like the A-argument. Given the role ranking $A/R > P/T$, we can thus formulate the still more general flagging universal in (38).

(38) The role-reference association universal (Haspelmath 2021b: 125)

Deviations from usual associations of role rank and referential prominence tend to be coded by longer grammatical forms if the coding is asymmetric.

Thus, flagging differs from indexing in that it is not always triggered by high referential prominence when it is differential: Only P and T arguments show special flagging when referentially prominent, while A and R show it when they are non-prominent. In the next section, we will see a plausible explanation for the tendencies of differential flagging.

5. Explaining differential flagging

There is a commonly expressed view according to which there are two competing functional motivations for differential flagging: a DISCRIMINATORY function (or “distinguishing”, “disambiguating” function), and an IDENTIFYING function (or “highlighting”, “indexing” function). Such a view has been expressed, for example, by Malchukov (2008: §4; 2017), Dalrymple & Nikolaeva (2011: 2), and Schikowski & Iemmolo (2015: §1.3). Witzlack-Makarevich & Seržant (2018: 30) put it as follows:

“The two most frequently mentioned functions of case marking here are the distinguishing function and the identifying function... The distinguishing function of case marking serves the purposes of disambiguation of the argument roles in clauses with two or more arguments. Case marking fulfills

¹⁴ Chappell & Verstraete (2019) express some doubts about this simplistic picture, and it may be that there is more that is going on (see also the example of differential subject flagging in nonfinite subordination in Turkish that is cited by Witzlack-Makarevich & Seržant 2018: 21). Even though there are not very many languages with differential subject flagging and perhaps occasional counterevidence, the generalizations in (37) do seem to be broadly on the right track (see also Schmidtke-Bode & Levshina 2018; as well as this 2019 blogpost <https://dlc.hypotheses.org/1768>).

the identifying function in that it codes the semantic relationship that the argument bears to its verb.”

However, it seems that this not the right dichotomy, because differential flagging is explained neither by the discriminatory nor by the identifying function. One problem with the disambiguation account is that differential object flagging is found also in cases where there is no subject, such as impersonals or even in passives (Irimia 2021). That discrimination (or disambiguation) is not the key factor also becomes clear when we consider still further phenomena, such as differential place marking (Haspelmath 2019a). For example, some languages have special place marking when the place is not inanimate, for example Italian, where the spatial preposition *a* is restricted to inanimate goal nominals, whereas animate goal nominals require the bisegmental preposition *da*.

(39) Italian

- a. *Vado a-lla chiesa.*
 I.go to-the church
 ‘I go to the church.’
- b. *Vado da-l poliziotto.*
 I.go to-the policeman
 ‘I walk up to the policeman.’

There is also differential zero-marking in some languages where no spatial preposition occurs when the place is expressed by a place-name. An example is Maltese, where the preposition *f-* ‘in’ does not occur with place names, e.g.

(40) Maltese (Stolz et al. 2017: 463)

- a. *Jgħallem Ghawdex.*
 3SG.M.IMPFV.teach Gozo
 ‘He teaches on Gozo (an island).’
- b. *Jgħallem f-l-iskejjel ta-l-Gvern.*
 3SG.M.IMPFV.teach in-DEF-schools of-DEF-government
 ‘He teaches in the schools of the government.’

Place-names are not less confusable with subjects than common place nouns, so these contrasts cannot be explained by the discriminatory function (see also Haspelmath 2026). Instead, what explains all cases of differential flagging is expectation sensitivity:

(41) **explanation in terms of expectation sensitivity**

Speakers add extra coding material when a meaning is unexpected in its context, and this may be grammaticalized.

For example, spatial meanings are expected with place nouns like ‘church’ or place names like ‘Gozo’, but less expected with human nouns or place nouns that are not names. And the patient meaning is more expected with indefinite and inanimate

nominals than with definite and animate nominals. More generally, asymmetric coding in grammar is usually due to coding efficiency and reflects a tradeoff between speaker needs and hearer needs (Haspelmath 2021a).

An explanation along these lines has long been advocated. Some relevant quotations from the 20th century and from the less well-known literature are given below.

“[...], the most **natural kind of transitive construction** is one where the A is high in animacy and definiteness, and the P is lower in animacy and definiteness; and any **deviation** from this pattern leads to a more **marked** construction.” (Comrie 1989: 128)

“**disagreement** of slot and filler properties favors markedness, **harmony** favors unmarkedness.” (Bossong 1991: 163)

“the need for overt coding of topical direct objects is not due to the fact that a topical direct object is cognitively more marked per se. Rather, I have argued that **it is the lower frequency of topical/topicworthy referents in direct object position** that explains why these direct objects are less easy to process. In addition, we have seen that language processing is based on the **expectations and predictability** of the language user. ... Since the most frequent situation seems to be one in which agents are human, definite and topical and direct objects are less human, definite and/topical than agents, the reversal of this situation would be likely to be harder to process.” (Iemmolo 2011: 268-269)

Summarizing the explanation, we can say that referentially prominent P-arguments tend to get extra flagging because P-arguments are usually non-prominent in language use, while referentially non-prominent A-arguments tend to get extra flagging because A-arguments are usually prominent. We do not need terms and concepts such as “markedness”, “naturalness”, or “iconicity”, which were prominent in much earlier work (e.g. Aissen 2003).

But of course, these are not the only explanations of the differential object marking tendencies that have been proposed. Bárány & Kalin (2020) provide an overview of some of the most commonly advanced ideas, also from the generative tradition. In generative syntax, one influential idea has been that objects are accusative-marked (and subjects not ergative-marked) when they “move up” to a higher position in the tree, because case assignment happens configurationally (e.g. Coon & Preminger 2017; Baker 2024). These approaches have been developed quite independently from the functional proposals discussed here, and they are rarely compared with them.¹⁵

6. Differential indexing is uniform across roles

We saw in §2-3 (on differential indexing) that the major roles of a transitive clause (A and P) behave similarly for indexing (DOI and DSI are favoured by referential prominence), and in §4 (on differential flagging) that the major roles behave

¹⁵ But see Haspelmath (2019c) for a brief comparison of the efficiency-based approach with Baker’s dependent-case approach.

differently. In the current section, we will briefly look at differential indexing in two additional roles: cross-indexing of the recipient and cross-indexing of the adnominal possessor.

Differential indexing has sometimes been described for the recipient argument of ditransitive clauses. In Ait Seghrouchen Berber, for example, definite recipients are cross-indexed, while indefinite recipients are not, as seen in (42).

(42) Ait Seghrouchen Berber (cited after Souag 2014: 1-2)

a. *Wši-x* =*as* *aysum* *i* *wmušš*.
give.PFV-1SG =3SG.DAT meat to cat
'I gave meat to the/*a cat'. (Guerssel 1995: 115)

b. *Wši-x* *aysum* *i* *wmušš*.
give.PFV-1SG meat to cat
'I gave meat to the/a cat'. (Guerssel 1995:115)

Similarly, in Macedonian, definite recipients are indexed (by *mu* in example (43a)), but indefinite recipients are not indexed (as in (43b)) when they are non-specific or indefinite.

(43) Macedonian (Mišeska Tomić 2008: 453-454)

a. *Jana mu go dade pismo-to na dete-to*.
Jana him.DAT it.ACC gave letter-DEF to child-DEF
'Jana gave the letter to the child.'

b. *Jana go dade pismo-to na edno dete*.
Jana it.ACC gave letter-DEF to a child
'Jana gave the letter to the child.'

The general preference for cross-indexing referentially prominent arguments seems to extend to adpossessionive indexing as well. For Modern Greek, Alexiadou (2016) reports that independent personal pronouns are obligatorily cross-indexed by possessive person indexes, while full nominal possessors cannot be indexed with *tu* (3SG possessive index):

(44) Modern Greek (Alexiadou 2016: 3)

a. *to vivlío =mu eména* (**to vivlío eména*)
the book =1SG me
'MY book'

b. *to vivlío tu fitití* (**to vivlío=tu tu fitití*)
the book the.GEN student.GEN
'the student's book'

Alexiadou also mentions a similar situation in French (*son livre à lui* 'HIS book' vs. **son livre à Jean* 'Jean's book'), and there are two further languages where a very similar pattern has been reported, Khanty (Uralic) and Welsh.

(45) Khanty (Nikolaeva 1999: 14; 52; cited by Siewierska 2004: 152)

a. *luw xo:t-əl-na*
 he house-3SG-LOC
 ‘in his house’

b. *Juwan xo:t-na*
 Juwan house-LOC
 ‘in Juwan’s house’

(46) Welsh (Roberts 1999: 622; cited by Siewierska 2004: 153)

a. *ei=wraig o*
 3SG=wife he
 ‘his wife’

b. *(*ei=wraig Gwyn*
 (3SG=wife Gwyn
 ‘Gwyn’s wife’

Summarizing, we have seen that differential indexing does not generally behave like differential flagging in that it does not show the mirror-image behaviour. Instead, it appears to generally conform to the prominent indexing generalization in (9). Linguists have not studied differential indexing in a fully systematic way so far, so the evidence for this claim is not very robust yet, but it is hoped that the present paper will stimulate further research.

How the differential indexing universal should be explained is a question that I will not address fully here. According to Siewierska & Bakker (2009: 293), indexing can be thought of “as a form of pronominalization used in preference to free personal pronouns for highly salient discourse referents, which the speaker assumes to be easily accessible to the addressee”. For this reason, it is expected that it will be found primarily with referentially prominent arguments. In the same vein, Just (2024: 314) notes that indexes often come from anaphoric pronouns but become “less referential” when they grammaticalize, but “it is the transmission of the prominence level of the referent which remains”. The basic idea has been around since Givón (1976), who noted that anaphoric pronouns can become “agreement markers” but continue to be used for tracking the reference of salience discourse participants.

7. That differential object indexing is different is not new

It is not a new insight that differential (object) indexing is different from differential (object) flagging in its patterning. The difference between DOF and DOI was clearly recognized, for example, by Iemmolo (2011). Before getting to the earlier literature on flagging vs. indexing, however, I will briefly review the history of the terms used for the phenomena.

The term *differential object marking* was introduced into the general literature by Bossong (1982; 1985; 1991) and became prominent at the latest with Aissen’s

(2003) well-known paper.¹⁶ However, the first occurrence of the expression “differential marking of objects” is in Moravcsik (1978: 275), and Moravcsik’s statement of the universal regularity is clearer than Bossong’s (1991):¹⁷

“If there is any semantic difference between an accusative marking and ... a nominative marking (to the exclusion of passivization), this semantic difference will be related either to definiteness, or to animacy, or humanness ..., with the accusative, by definition, marking the more definite (rather than the less definite), the animate or human (rather than the inanimate or non-human), ... noun phrase.” (Moravcsik 1978: 283)¹⁸

Since Aissen (2003) and de Hoop & de Swart (2009), the corresponding term *differential subject marking* has also been known quite widely. Alternatively, some authors started using *differential P marking* and *differential A marking*, as it was more and more recognized that the marking concerned a type of role (defined as a comparative concept, Haspelmath 2011), not a structural type of argument. In the present paper, and quite commonly in the literature, “object/P marking” and “subject/A marking” are used interchangeably (but see n. 7 on subject vs. A-argument).

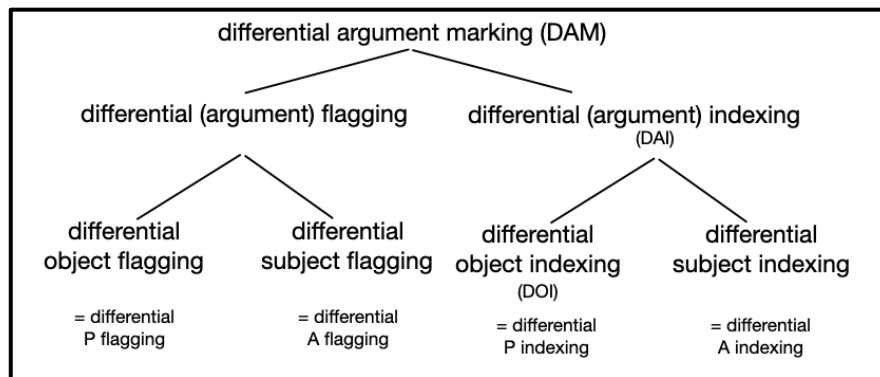
For the first two decades, “differential object marking” referred only to flagging (i.e. DOF), but Morimoto (2002) introduced differential object indexing into the discussion, adopting the same general explanatory approach as Aissen (2003). Before Iemmolo (2011 introduced “differential object indexing” (see n. 3), such phenomena had been discussed under the label of “clitic doubling” in Romance languages and elsewhere for quite some time (e.g. Steriade 1980), but it was only more recently that clitic doubling and prepositional differential object marking were brought together (e.g. Kallulli 2016, Bonfim Duarte 2023).

As a cover term for differential object marking and differential subject marking, there is now a term *differential argument marking* (e.g. Arkadiev 2009; Keine & Müller 2010; Witzlack-Makarevich & Seržant 2018), and as subtypes of DAM, one can distinguish *differential argument indexing* (DAI, Walker et al. 2024) and *differential argument flagging* (DAF). The various terminological distinctions are summarized in Figure 1.

¹⁶ See Bossong (2021) for a personal account, in which he highlights the role of his interactions with Gilbert Lazard in the 1970s (see Lazard 1978; 2001).

¹⁷ As far as I can see, Bossong did not make explicit universal statements in his 1982 and 1985 works, though the universal scope of his discussions is clear from the context.

¹⁸ See also this blogpost for further discussion: <https://dlc.hypotheses.org/2764>

Figure 1. The terminology of differential argument marking

We saw in §1 above that a number of authors have proposed that both subtypes of DAM (differential flagging and differential indexing) are instances of the same phenomenon, but it was recognized clearly by Iemmolo (2011) that the two subtypes have different motivations. While differential argument flagging is motivated by predictability and efficient coding (form-frequency correspondence, Haspelmath 2021a), differential flagging is motivated by topicality:

“Overt coding on topical direct objects is indeed motivated by the unexpectedness of highly topical/topic-worthy lexical direct objects, because of the low frequency of such NPs in direct object position. By contrast, indexation is correlated with a high degree of topicality of the referent it indexes. In other words, overt coding serves to signal that a lexical NP fulfilling the role of direct object is highly topical, while indexation is naturally associated with topical NPs and signals the discourse prominence of such direct objects.” (Iemmolo 2011: 268)

Very similarly, Witzlack-Makarevich & Seržant (2018: 25-26) note (see also Croft 2022: 122):

“While agreement or indexing is “a topic related phenomenon” as Givón (1976: 185) puts it (cf. also Kibrik 2011), flagging is not related to topichood or information-structure in general, but rather to semantic argument roles and various dependency relations between a head and its dependent.”

But we can go back even further: The Prominent indexing universal (that we saw in (9) above) was formulated clearly by Siewierska (2004: 149), who used the term “person agreement” for what we now call *indexing*:¹⁹

(47) Siewierska’s indexing universal

“If person agreement is not obligatory in a language, it will occur with controllers displaying the characteristics on the left-hand side of the hierarchies [of referential prominence; see (7)-(8) above]”

¹⁹

For a recent comprehensive typological study, see Walker (2024).

Similar ideas had already been expressed by Croft (1988), and as early as 1979, Comrie argued against conflating differential flagging (“case marking”) and differential indexing (“agreement”). The key passages are printed in bold in the following quotation (again, “verb agreement” is used for indexing):

“Thus we can say, more generally, that **verbs tend to agree with those arguments that are, or at least: are typically, definite, animate, and thematic**. One might therefore be tempted to subsume both case-marking and verb-agreement under the same unified general explanation. We suggest that this cannot be done, although the fact that subjects tend to be definite, animate and thematic is no doubt an integral part of the explanation for both phenomena. The crucial difference, however, lies in the *discriminatory* role of case-marking of definite/animate objects: they are marked differently from subjects, although they share certain semantic features in common, indeed *because* they share certain features in common, as the function of this case-marking is to keep them apart overtly. **Verb-agreement, however, has little or no discriminatory function, indeed it serves to unite definite/animate direct objects and subjects.**” (Comrie 1979: 20)

Thus, the key observation of Just (2024) and of the present paper, that flagging and indexing behave differently in differential situations, is not new, but it is still worth highlighting, because DOM of both types (DOF and DOI) is so prominent in the Romance languages, and DSM patterns of both types (DSF and DSI) are less common and less uniform and therefore less widely known.

8. Conclusion

I have argued in this paper that differential object marking (DOM) is not a unitary phenomenon in the sense that its two subtypes, differential object flagging (DOF) and differential object indexing (DOI), probably have rather different explanations.

That different indexing is typically conditioned by referential prominence is well known from the earlier literature (especially Siewierska 2004: 149), and that the two types of different core argument flagging are sensitive to the opposite ends of the referential scales has also been said repeatedly. Thus, the main point of this paper is not new, but it seems worth emphasizing that this implies that there cannot be a uniform explanation for both differential flagging and differential indexing (as Aissen 2003: 474 was still hoping). As Just (2024: 311) notes, “indexing is primarily connected to prominence, irrespective of the argument role”. But flagging occurs primarily in order to counter expectations about the role that a hearer may have on the basis of an argument’s properties, so it occurs on P-arguments and T-arguments primarily when they are prominent, but on A-arguments and R-arguments primarily when they are non-prominent.

As always in assessing universalist approaches, it must be remembered that the claims made here pertain to the level of human language in general, and need not have direct consequences for language-particular studies. Over the last two decades, more and more works have examined the precise conditions under which variable differential argument marking occurs in particular languages, using sophisticated corpus evidence (e.g. Coghill 2014; Riesberg 2018; Walker et al. 2024) or

experimental evidence (e.g. Vega Vilanova & Fischer 2025; Wall et al. 2025). It has also been observed, for example, that object clitic doubling and object agreement by affixes obeys different language-particular regularities in some languages, as a reviewer observes. There are no direct consequences of my universalist results for language-particular findings, though ultimately one hopes, of course, that general and particular linguistics will inspire and support each other.

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